BU5300 – IT Project Management

Lecture 2: Business Case and Organizational Framework

Jefferson Seide Molléri

Høyskolen Kristiania

Agenda for today’s lecture

- Quick review of Lecture 1
- Chapter 3: The Business Case
- Chapter 4: The Organizational Framework

Takeaways from Lecture 1

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- What is a project?
- What is project management?
- Role of the project manager
- Work modes in organizations
- Project lifecycle
- The project constraints
- Traditional project management
- Agile project management
- The types of IS projects

Last week's assignment

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- Form a group (6–8 students) and create a Project Group in Canvas
- Brainstorm an **IT Project Idea**
  - Check the case example in Canvas: *Smart Tourist Navigation for VisitOslo*
- Detail the following:
  - A summary of the project (1-2 paragraphs)
  - The type of the project (which category of the 9 nine categories)
  - Potential strengths, weaknesses, opportunities and threats (SWOT analysis)
  - What type of technical skills are needed for such a project
- **41 unique submissions**

Last week's assignment
Type of Project and Technical Skills

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- **Software development**
- System enhancement
- Infrastructure implementation
- **Disaster recovery?**
- **Outsourcing/insourcing?**

- **Technical skills** (Front-end and back-end Developers, DB Manager, UX/UI Designer, Testers / QA, Cybersecurity Expert, etc)
- Subject Experts (geolocation, AI/ML)
- DevOps
- Project Management
- **Data Analysis?**
- **Legal & Compliance?**
- **System Integration?**

Last week's assignment
SWOT Analysis

Strengths
- Clear need (growing market)
- Potential local users
- Business experts inside

Weaknesses
- Requires internet connection
- Limited to Oslo
- Depends on the APIs

Opportunities
- Foster tourism business in Oslo
- Competence growth in organization
- First of a kind in Norway

Threats
- Competitors
- Security (sensitive information)
- Technology changes

Overview of the Course
Lectures and Book Chapters

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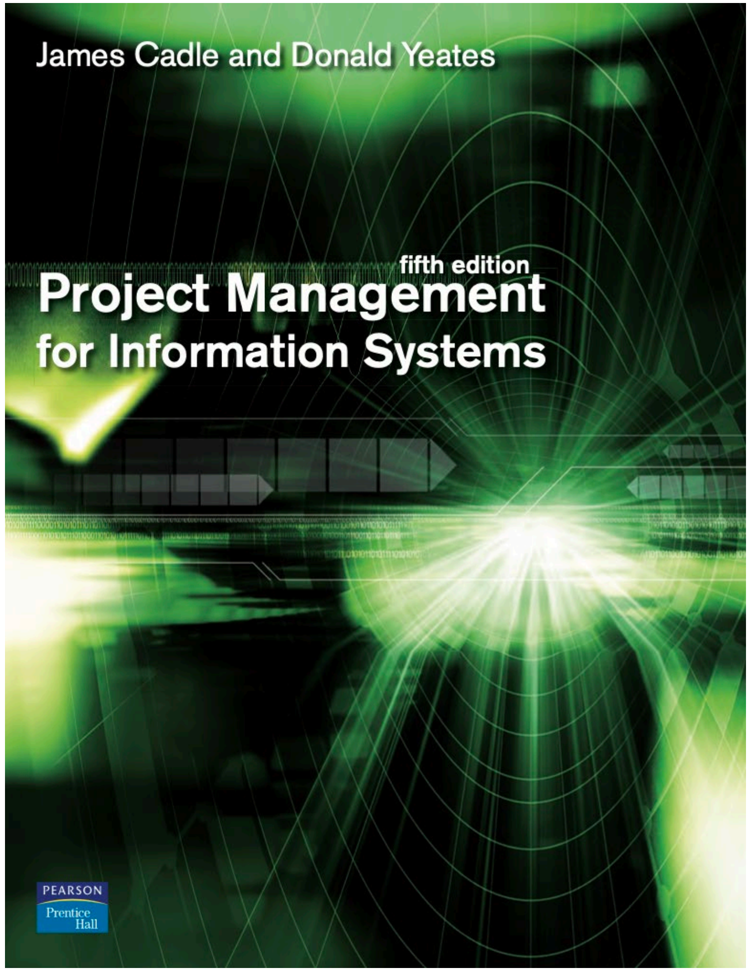

Lecture 1 - Chapters 1 and 2

Lecture 2 - Chapters 3 and 4

Lecture 3 - Chapter 6

Lecture 4 - Chapters 7 and 8

Lecture 5 - *Agile Development*

Lecture 6 - Chapter 9 and 10

Lecture 7 - Chapters 11, 12 and 13

Lecture 8 - Chapters 14 and 15

Lecture 9 - Chapters 18 and 21

Lecture 10 - Chapters 20, 22 and 23

IMPORTANT! Changes may occur. Always check on Canvas and TimeEdit

Chapter 3: the Business Case

Building the Business case for a project to obtain funding for it from the senior management

Learning outcomes

- Describe the structure of a business case
- Use the payback, discounted cash flow and the internal rate of return methods to calculate the financial implications of a business case
- Differentiate between tangible and intangible costs and benefits

What is Business Case?

A business case is a project management document explaining how a project's benefits outweigh its costs and why it should be executed. Business cases are prepared during the project's initial phase, and the goal is to include all the project's objectives, costs, and benefits to convince stakeholders of its value.

Business Case Examples
https://chasegroup.com.au/business-case-examples

- New Product Business Cases
- Business Cases for Equity Investments
- Software as a Service Business Cases
- Capital Equipment Business Cases
- Program Office Business Cases
- Start-up Business Cases
- Marketing Campaign Business Cases
- Sustainability and CSR Initiatives Business Cases
- Expansion into New Markets Business Cases
- Employee Training and Development Business Cases
- Customer Experience Enhancement Business Cases
- Infrastructure Business Cases
- AI Business Case Generator
- How to Build the Business Case

Introduction

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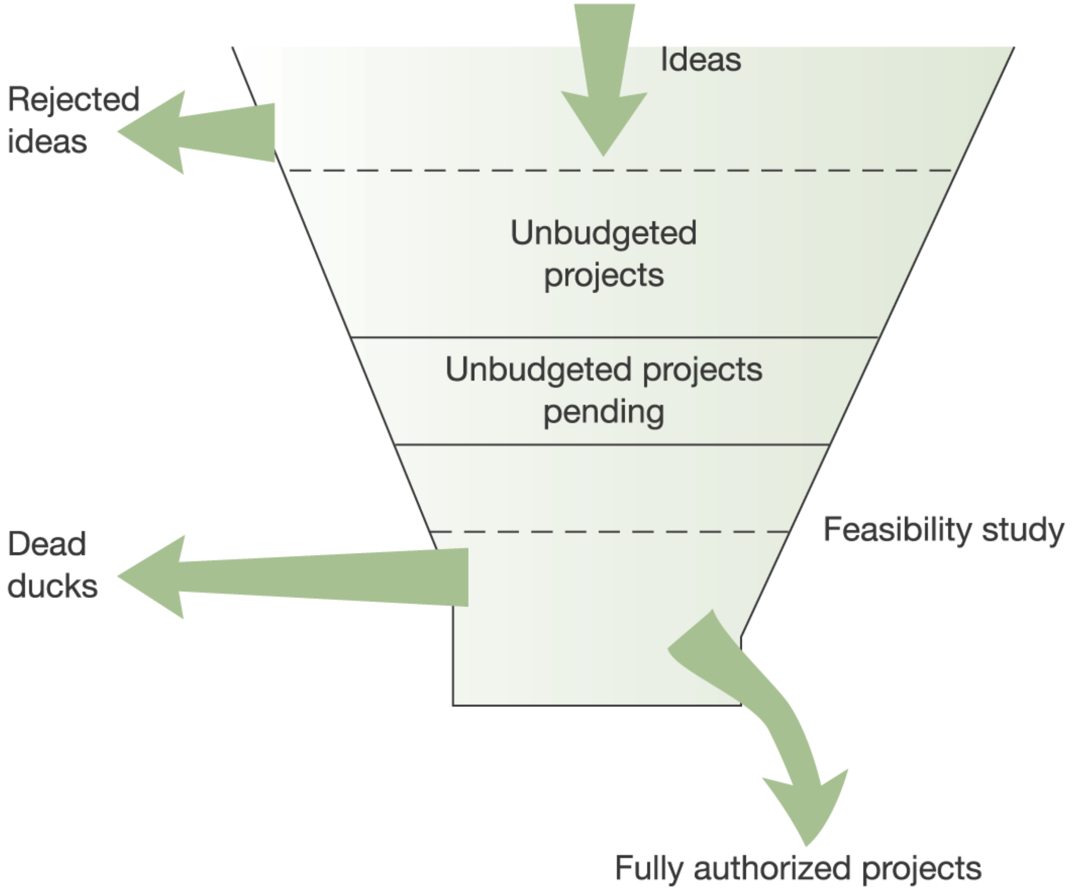
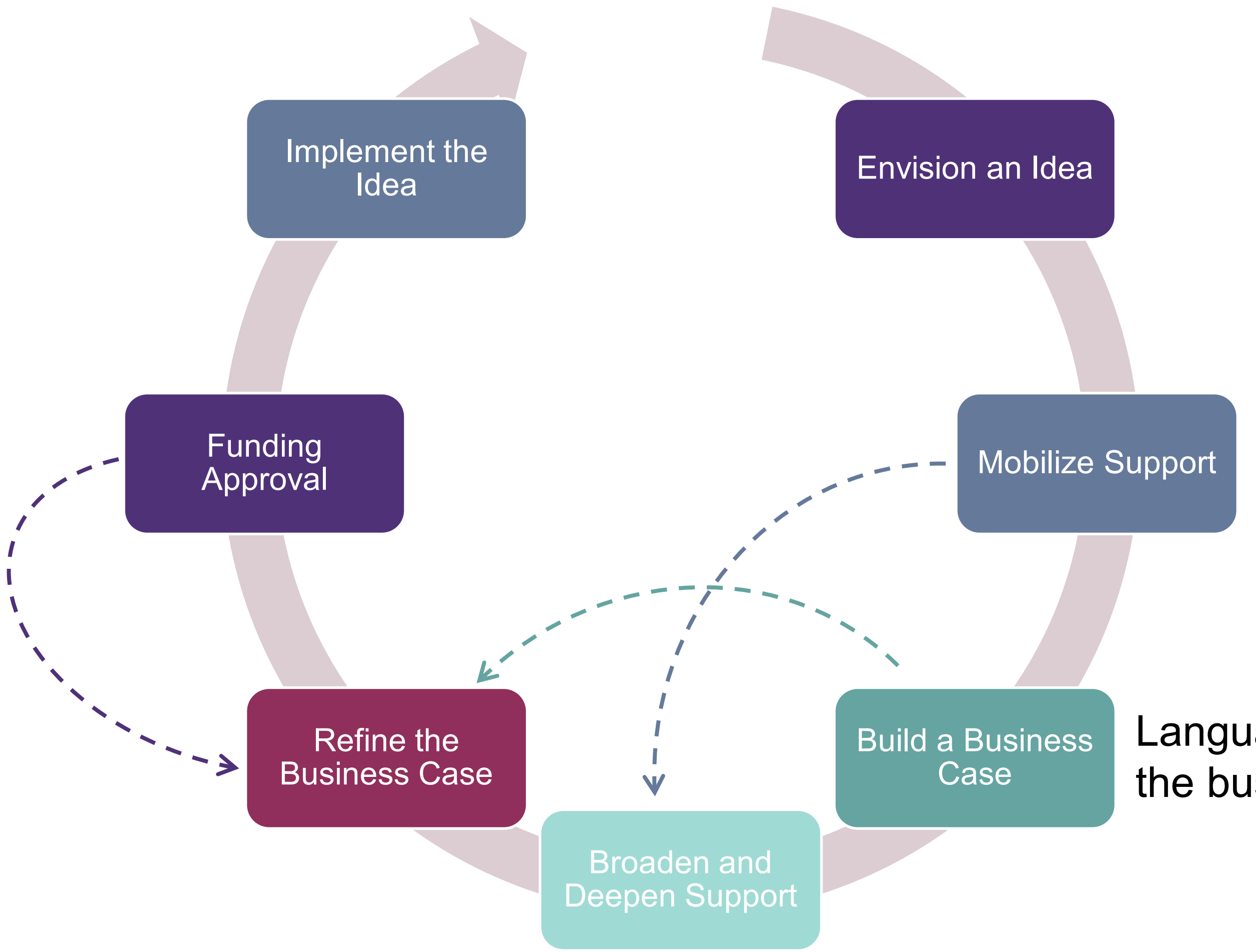
- The role of a business case in justifying the project
  - The What, Why, Cost, and Timeline
- Why → Project benefits



The Innovation Cycle

from envision to implement an idea



Engage stakeholders early

Language of the business

Sidenote:
Starting a Business

Examples from https://www.innovasjonnorge.no/

Funding a Start-up

Turning your idea into a product

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- **Market Study**
  - Identify a problem or need
  - Develop a prototype (or MVP)
  - Test it in the market, demo to customer
  - Develop a business model
- Subsidy for market clarification
  - Up to 100 KNOK
- **Getting a sponsorship**
  - Establish a company
  - Further develop the idea
  - Introduction to the market
  - Follow-up and guidance in the first years
- Grants for commercialization
  - Up to 700 KNOK
- Start-up loan
  - From 500 KNOK to 1.5 MNOK

Grants for Student Entrepreneurship

Students' ideas into business

- **A new grant focused on newly-graduated students**
  - Start-up companies with market acceptance and growth potential
  - In cooperation with university and a commercial mentor
- Application deadline 29 Feb 2024
  - Webinar for applicants 11 January
- Up to 1 MNOK (3-5 years sponsorship)
- Check more here!

Content of a business case

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Project Business Case Example			
Project Name	Sales Team IVR Telephone System		
Project Sponsor	Head of Sales	Project Manager	Name of project manager
Date of Project Approval	3rd March	Last Revision Date	3rd March
Contribution to Business Strategy	Our strategy is to project best in industry customer service, and the current situation does not reflect this. The new IVR system will ensure all calls are answered in a timely manner. It will also ensure that calls are delt with efficiently. These two facts align this project to the company strategy.		
Options Considered	Options considered included: 1. Adding additional staff to sales team 2. Having a dedicated team for our best customers 3. An IVR system (selected)		
Benefits	1. Increased sales - currently extimated we lose 4% of all sales calls due to current issues. 2. Happier customers - we estimate new customer satisfaction will increase by 10%. 3. Improved LTV - lifetime value of customers will increase by 5% due to the two points above		
Timescales	Initial analysis shows that the system will take approximately 3-4 months to implement.		
Costs	IVR software = $35,000 Project Management = $30,000 Software team of 3 for 3 months = $90,000 Total estimated cost = $155,000		
Expected Return on Investment	Year 1 = $0 Year 2 = $120,000 Year 3 = $180,000 as LTV begins to be felt.		
Risks	Right now the project looks pretty straightforward but there are still some unknows surrounding implementation. There is also the risk that the project doesn't meet the sales team or customers needs. For this reason it is recommended to involve the sales team closely.		

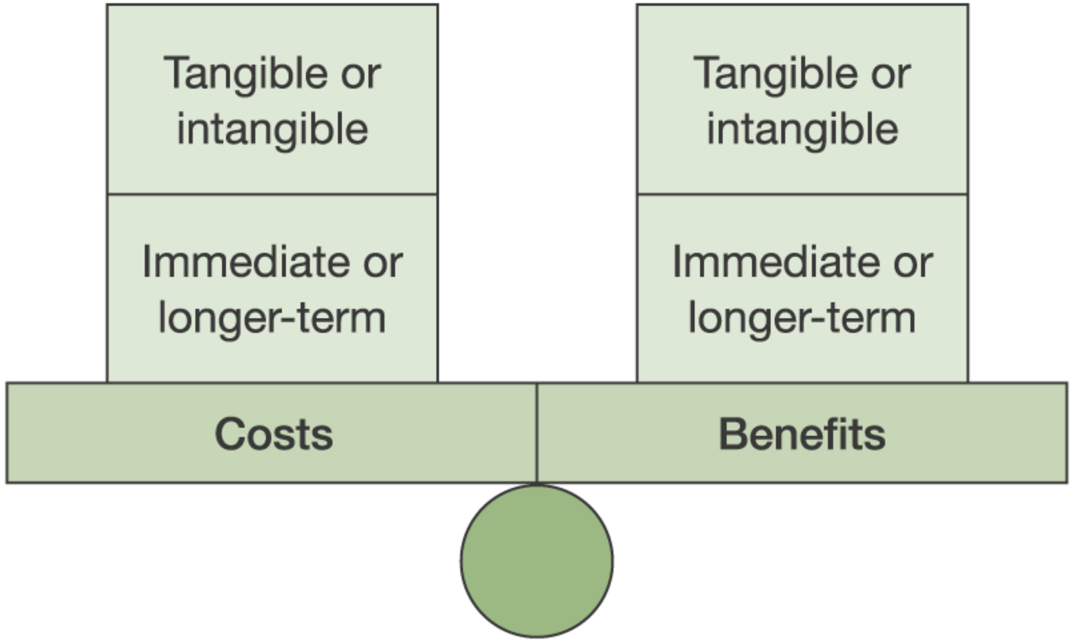
- Introduction and background
- Management summary
  - Description of problem or opportunity
  - Options available and considered
- Cost/benefit analysis
- Impacts and risks
- Conclusions and recommendations

Management summary

A critical section, especially in large business cases

- A statement of what is the problem or opportunity that the project is intended to address
- A list of the options considered and why those not chosen have not been recommended
  - Including the option of doing nothing (and the risk associated)
- A statement showing which option has been recommended, why, and what business benefits are expected to flow from it

Costs and benefits



- **Tangible** (which means they can be plausibly quantified in some way) or
- **Intangible** (which means that they cannot be so quantified)
- **Immediate / long term** realization

IT Costs

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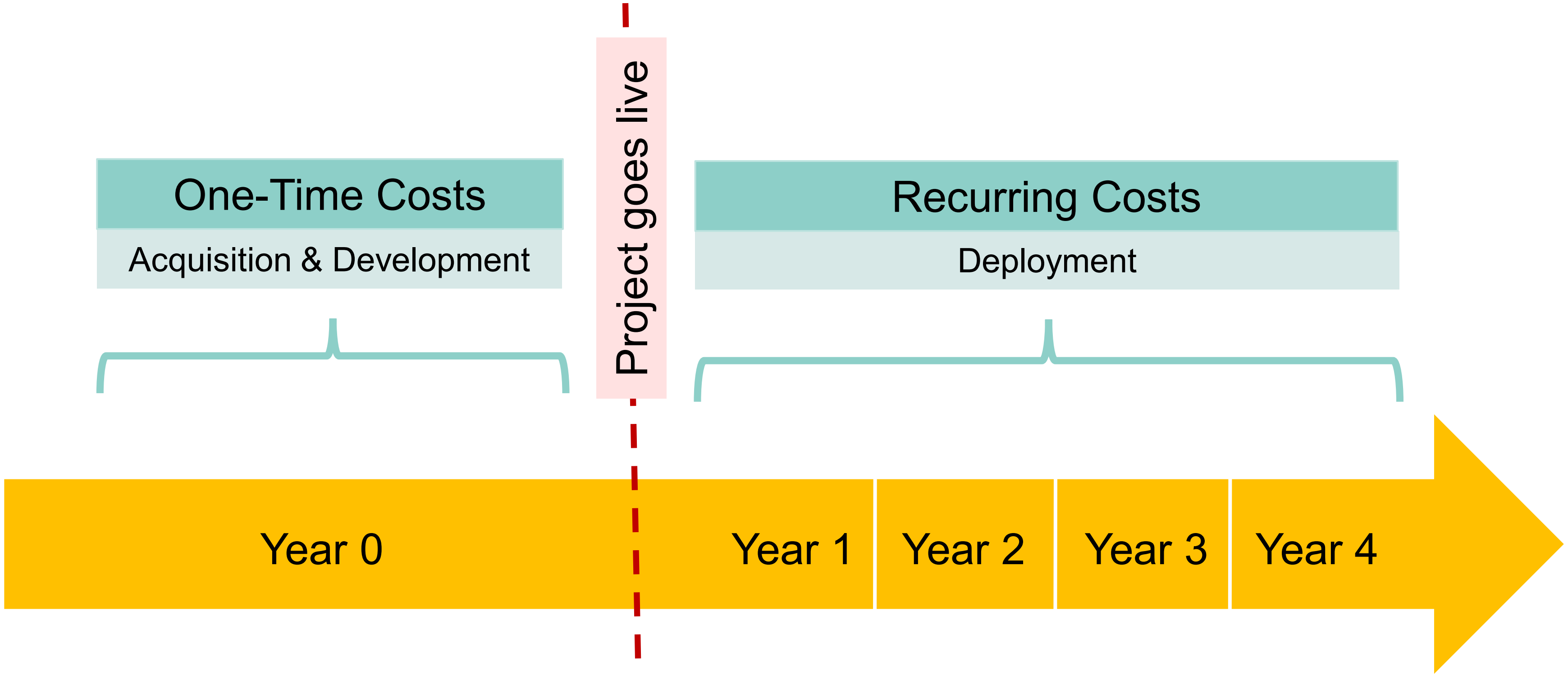
- List all costs
  - Don't lose credibility by not mentioning some
- One-Time versus recurring
- Tangible versus Intangible
- Central Site and Local Sites
- Timing of Flows

One-Time Costs
Initial purchases of hardware and software
Salaries of IT, employees, business employees and consultants involved in IT-related/work-related design, development and installation activities
Site preparation costs
Pre-installation training costs
Work disruptions during design, development and installation

Recurring Costs
IT infrastructure costs involved in operating the installed IT-enabled business solution
Salaries of IT employees, business employees and consultants involved in maintenance or enhancement activities
Hardware/software license fees, maintenance fees, and upgrades
Post-installation training costs
Work disruptions associated with: • Employees learning new work practices • Maintenance or technical enhancement activities

IT Costs
one-time vs recurring



Sidenote: Funding plan Innovasjon Norge

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Benefits Associated with Proposed IT Investments

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Benefit Type	Examples
Cost Reduction	• Lower current staffing levels • Lower current cost-of-goods-sold • Lower current inventory costs • Lower current cost of storing a gigabyte of data • Lower lost sales due to stockouts (a current cost of doing business)
Cost Avoidance	• No need to hire the additional staff (whose funding has already been approved) to handle expected increases in work volume • No need to build a planned new distribution center
Value Enhancement	• Higher gross margin (with stable cost structures) • Higher rate of sales growth • Higher rate of repeat sales • Higher rate of new product introduction • Higher rate of product/service improvement

Investment appraisal methods

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- Payback calculation – simple and straightforward
- Discounted cash flow:
  - Takes into account 'time value of money'
  - Produces a net present value for project
  - Sensitivity analysis can be applied
- Internal rate of return – produces single headline number for comparisons

Payback calculation

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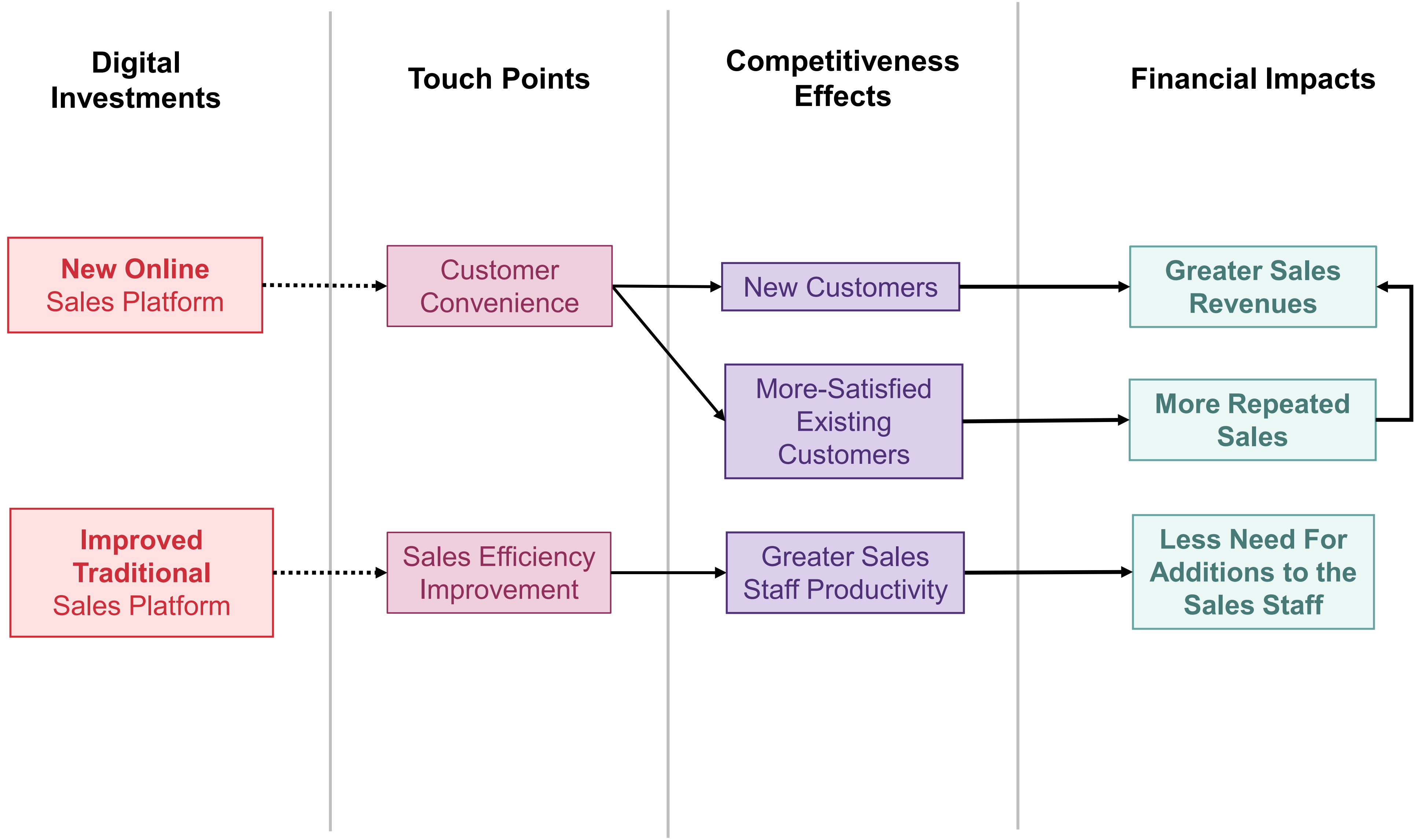
Item	*Year 1*	*Year 2*	*Year 3*	*Year 4*	*Year 5*
Hardware purchase	500,000				
Hardware maintenance	50,000	50,000	50,000	50,000	50,000
Software purchase	180,000				
Software support	20,000	20,000	20,000	20,000	20,000
Cumulative total costs	750,000	820,000	890,000	960,000	1,030,000
Staff savings per year	220,000	220,000	220,000	220,000	220,000
Cumulative savings	220,000	440,000	660,000	880,000	1,100,000
Cumulative savings less costs	–530,000	–380,000	–230,000	–80,000	+70,000

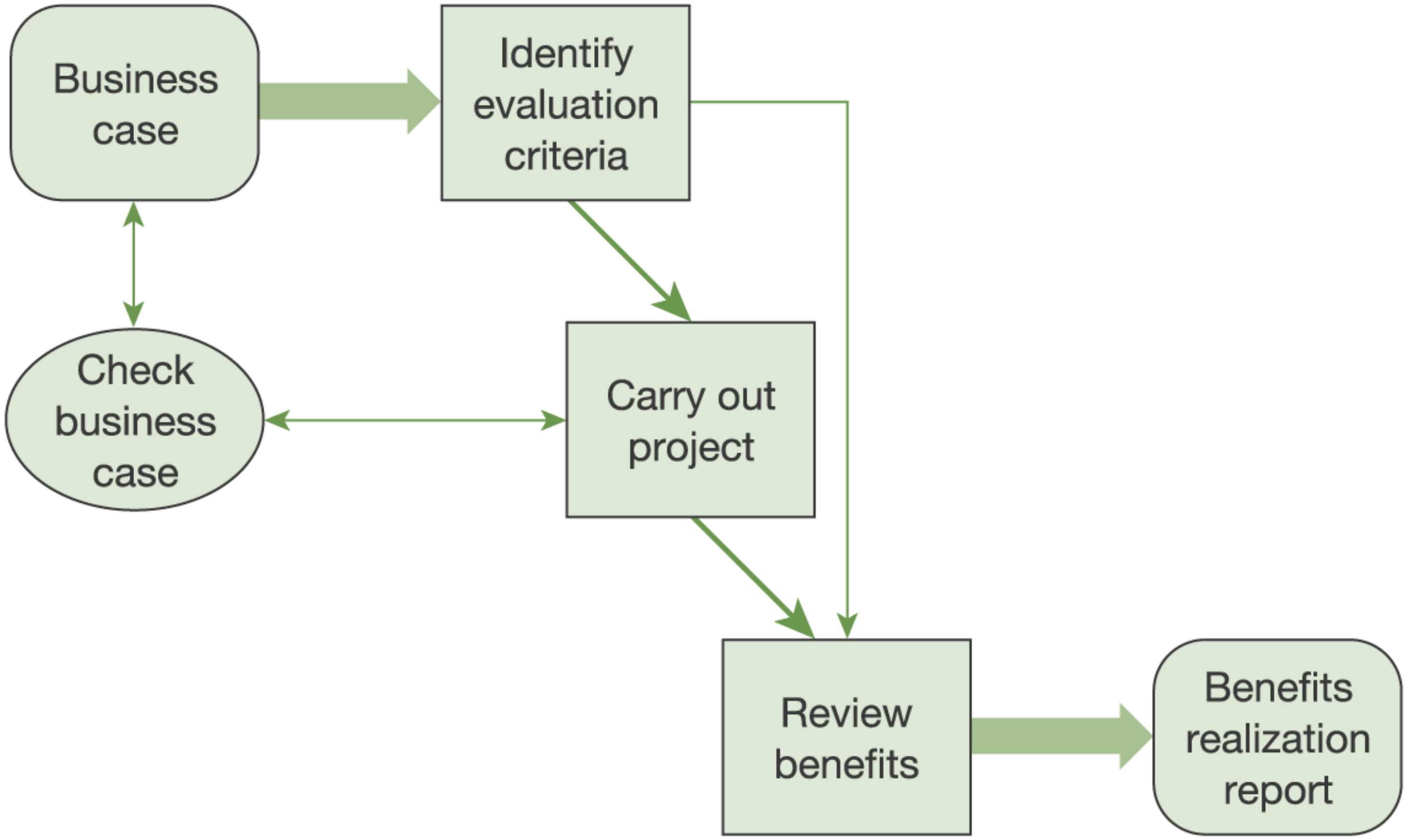
Net present value calculation

Year	Cash-flow for year	Discount factor at 20%	Discounted cash flow
1	–530,000	1000	–530,000
2	+150,000	0833	124,950
3	+150,000	0694	104,100
4	+150,000	0579	86,850
5	+150,000	0482	72,300
Net present value			**–141,000**

Cause and Effect Tool



Benefits realization



Presenting the business case

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- **Aim:** what is this designed to achieve?
- **Audience:** who is making the decision?
- **Arrangement:** logical, leading to the recommendation
- **Appearance:** accessible, attractive, easy to follow

Chapter 4: The Organizational Framework

Learning outcomes

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- Prepare organizational structure charts for a functional organization structure, a project structure and a matrix structure
- Identify the key roles and responsibilities in an IS project
- Define programme and project management

Functional organization

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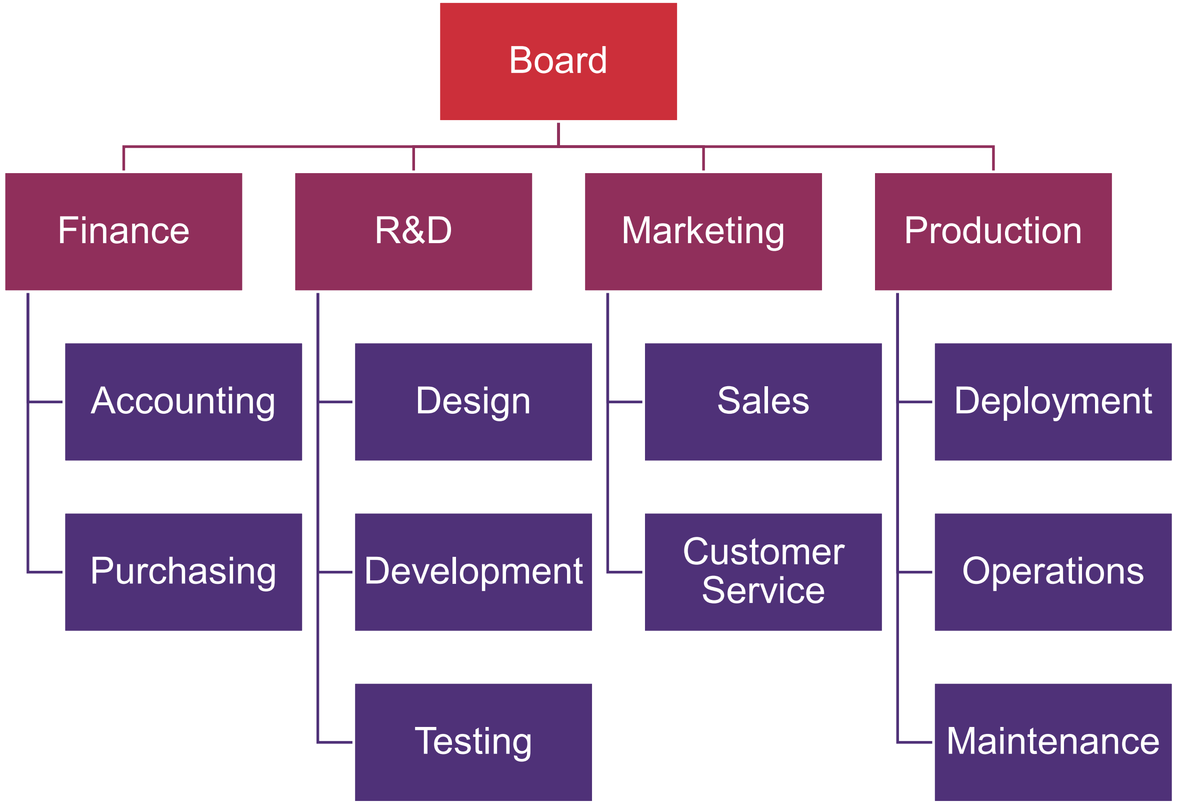
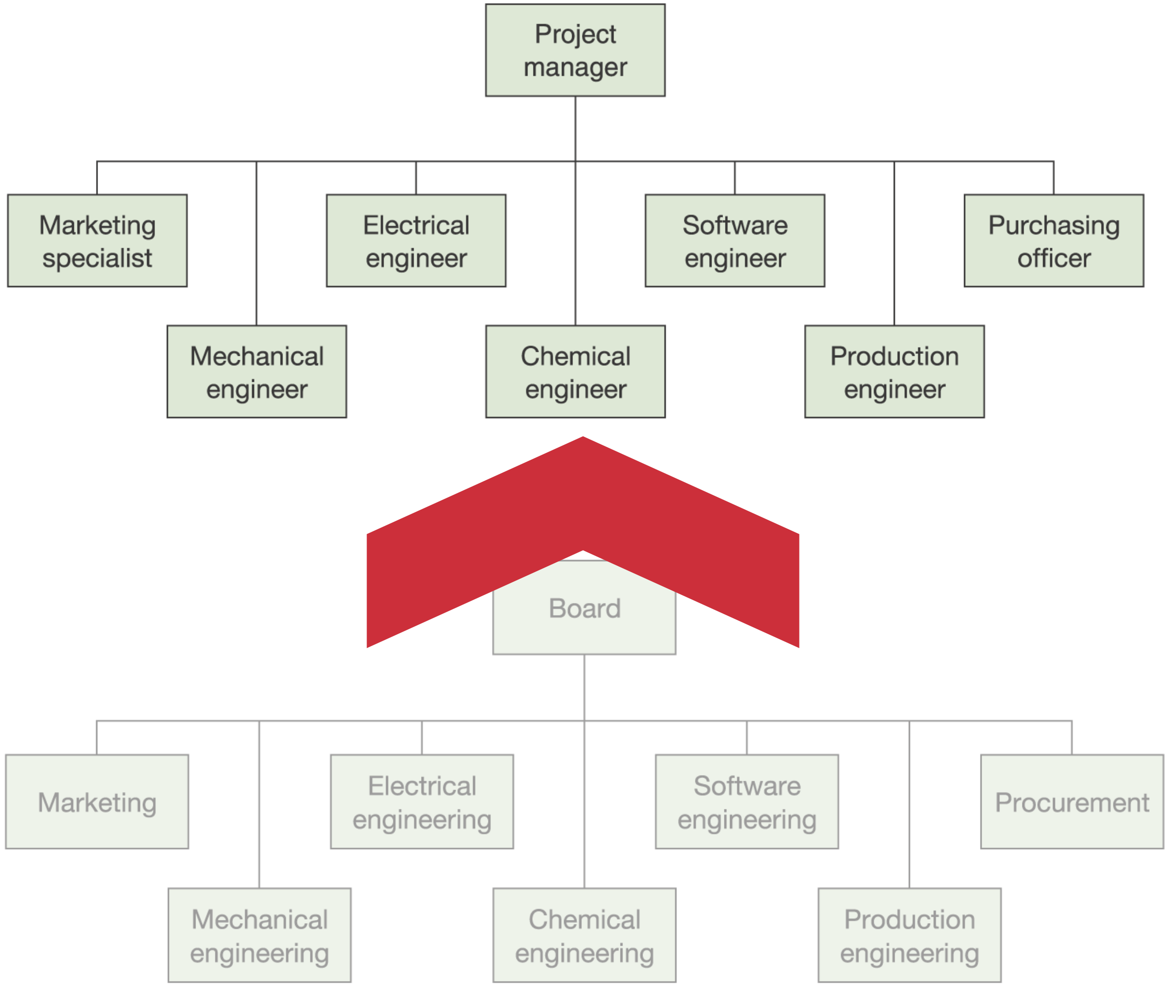


- The organization is divided into departments based on specialization
- Advantages:
  - Encourage the development of a high degree of specialist skills
  - People in the functions report to managers who understand their fields
- Disadvantages
  - can develop a ‘**silo**’ mentality – being more concerned with departmental objectives than with the needs of the organization as a whole

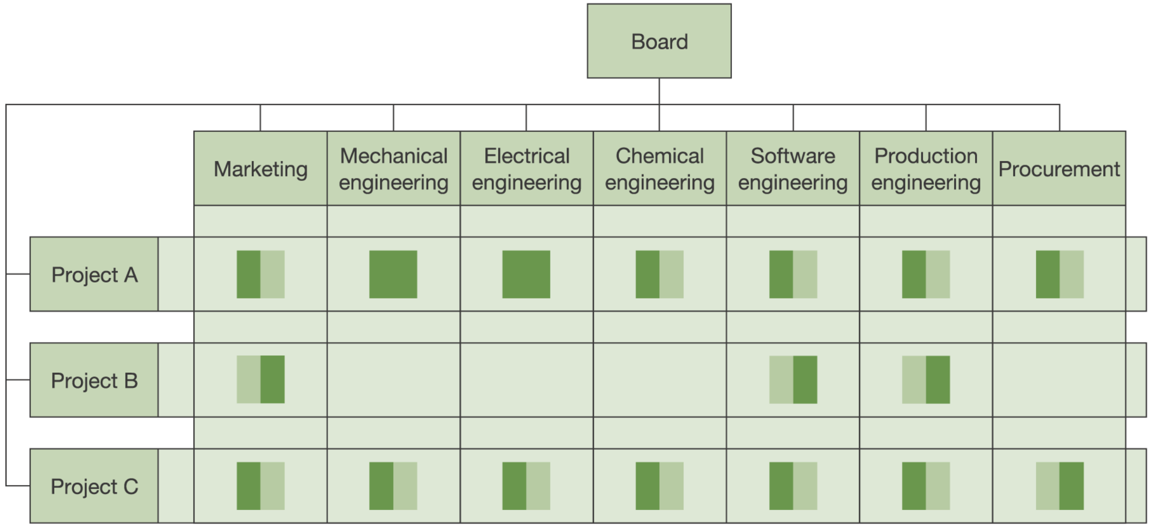
‘Pure’ project structure



- Set up projects organizations to carry out specific tasks
- Team members are drawn from across the functions
- Once the project is over, everyone goes back to their ‘home’ departments, or another project

The Matrix Structure

- People have more than one boss
  - To the head of their function (as far as professional issues are concerned), and
  - To a project manager for project-related issues
- More resource-efficient than the pure project since resources can be shared between projects



Generic project organization and roles

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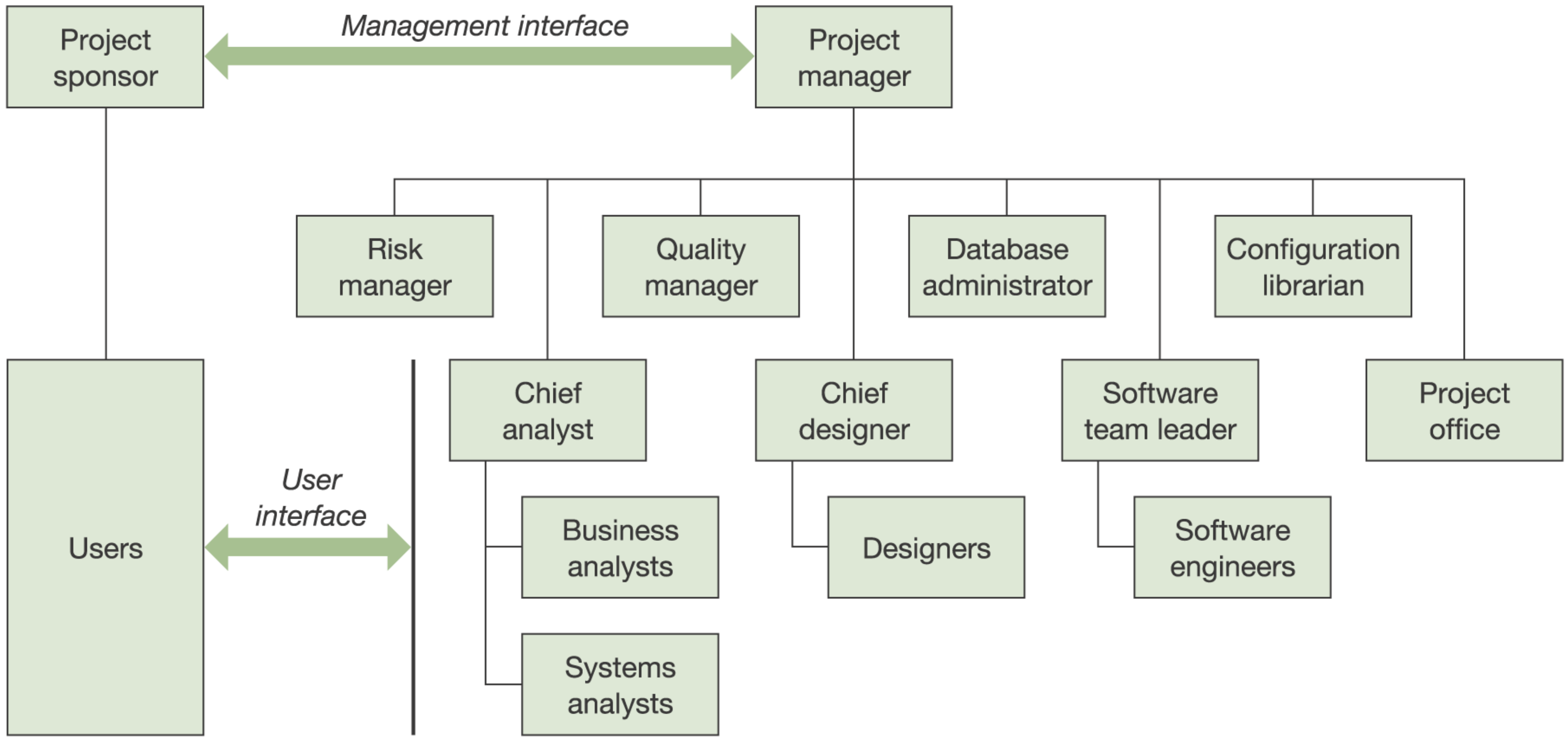
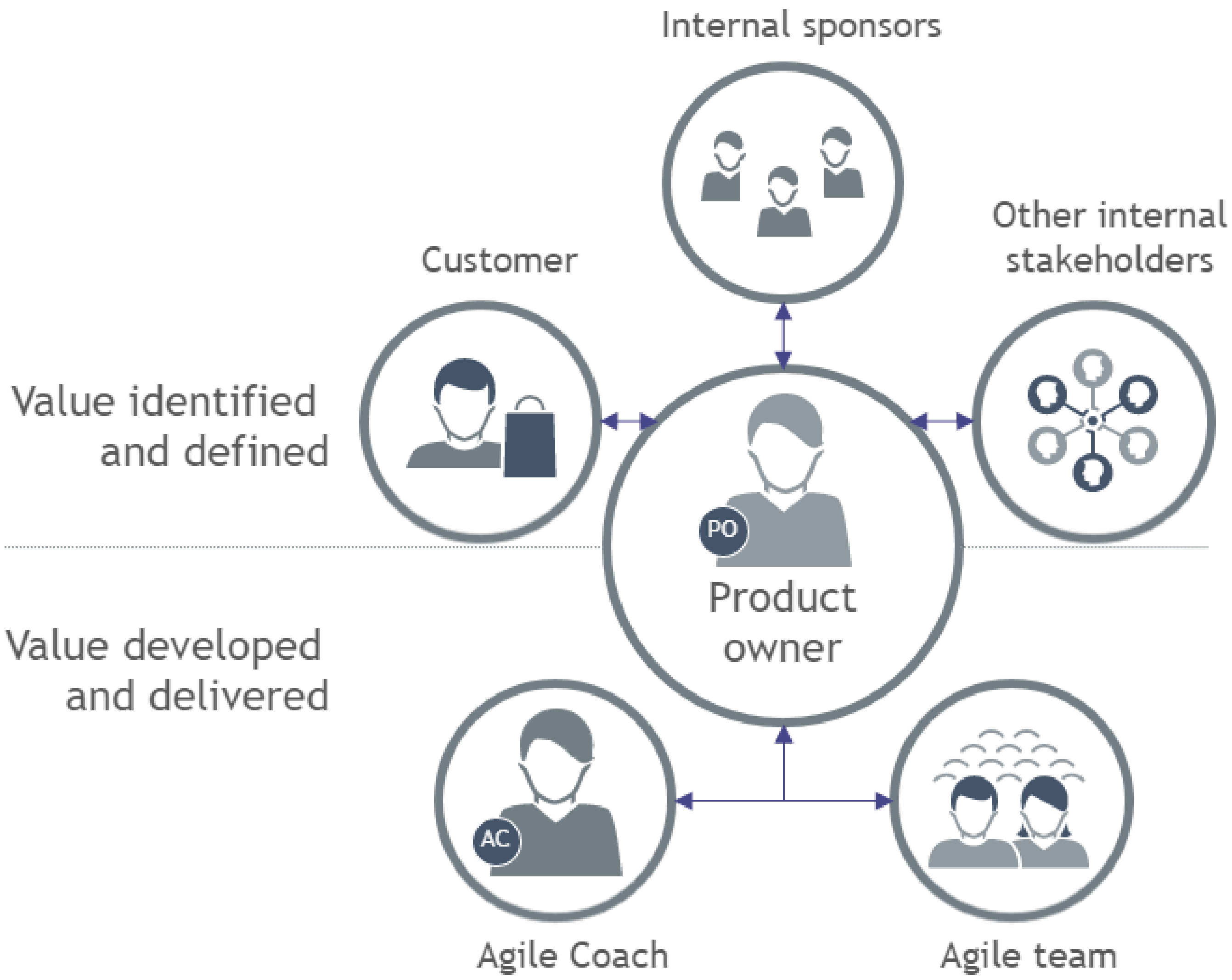


(Agile) organization and roles



The Role of the Sponsor

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- Define the business aims of the project
- Justify the project to the board
- Define the project's objectives and its priorities in terms of the 'triple constraint'
- Specify the minimum requirements that the project must meet if it is to achieve its business objectives
- Obtain approval for any capital expenditure involved
- Initiate the project and appoint the project manager
- Monitor the progress of the project from a business standpoint
- Monitor also the business environment to ensure that the project still meets the business needs
- Keep the board or higher management informed of progress
- If necessary, terminate the project
- Accountable for the success of the investment
- Provide high-level support as a champion for the project

The Role of the Project Manager

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- Achieve the project's objectives within the time, cost and quality/performance constraints imposed by the sponsor
- Make or force timely decisions to assure the project's success
- Plan, monitor and control the project through to completion
- Select, build and motivate the project team
- Keep the sponsor and senior management informed of progress and alert them to problems
- Recommend termination of the project to the sponsor, if necessary
- Serve as the principal point of contact between the sponsor, management and contributors
- Select and manage subcontractors

Questions?

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Course Activity 2

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Build a **Business Case document (or presentation)** for your IS project idea

- Introduction and background
- Management summary
  - Description of problem or opportunity
  - Options available and considered
- Cost/benefit analysis
- Impacts and risks
- Conclusions and recommendations

Tip! You can use the PRINCE's template available at:
https://kristiania.instructure.com/files/1054531

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